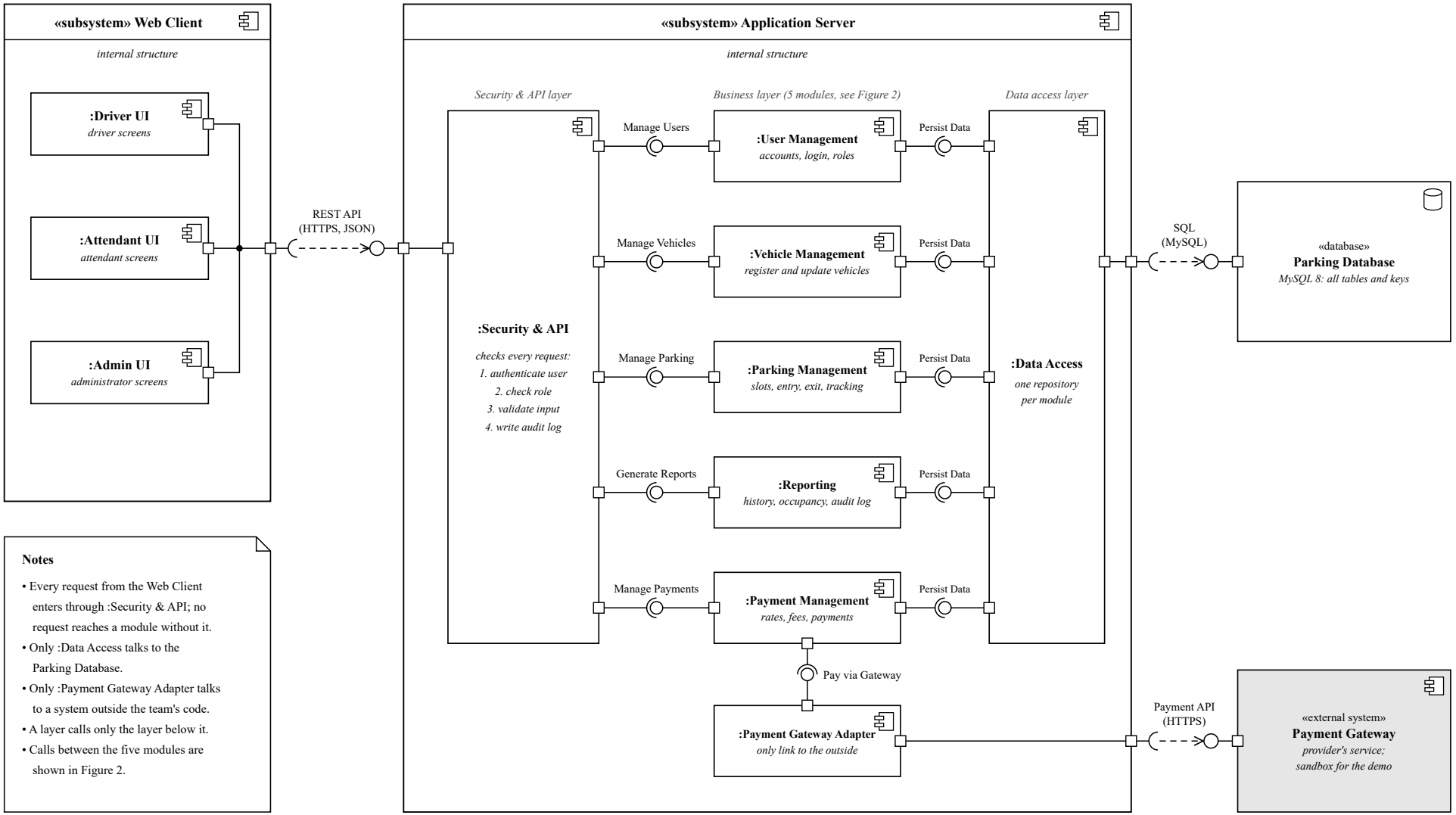
Vehicle Parking Management System

Figure 1 – Architecture: UML component diagram (subsystems and their internal components)



Notes

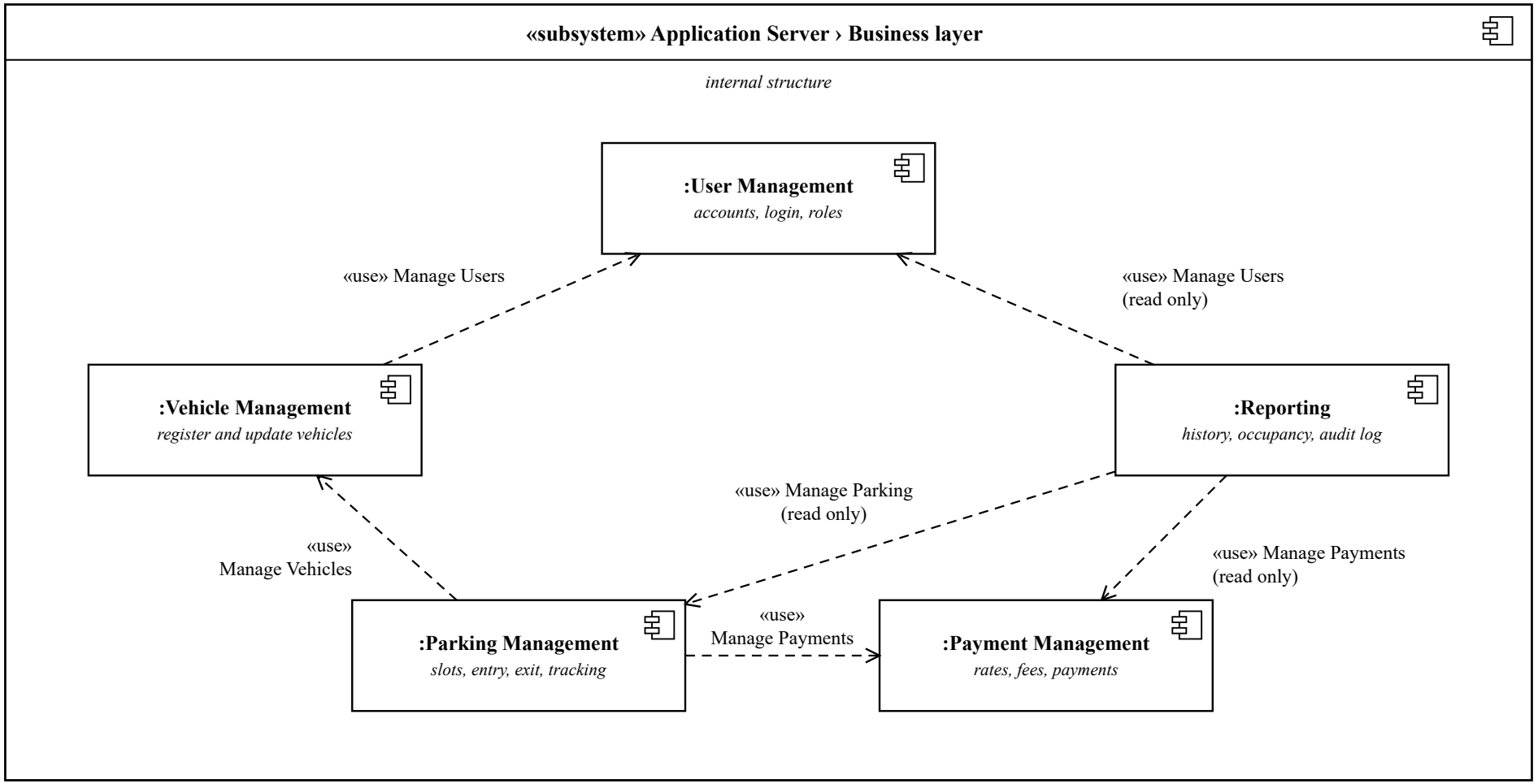
- Every request from the Web Client enters through :Security & API; no request reaches a module without it.
- Only :Data Access talks to the Parking Database.
- Only :Payment Gateway Adapter talks to a system outside the team's code.
- A layer calls only the layer below it.
- Calls between the five modules are shown in Figure 2.

Legend

- Component: a part of the system the team builds
- «subsystem»: components that run together as one process
- «external system»: not built by the team
- Port: where a connection enters or leaves a box
- Provided interface (ball): a service the part offers
- Required interface (socket): a service the part needs
- Solid line: delegation, a port passes calls to an inner part
- Ball in socket: the socket side calls the ball side
- Dashed arrow: call between subsystems, points to provider

Vehicle Parking Management System

Figure 2 – Business layer: dependencies between the five modules (inside :Application Server)



Notes

- Dependencies go one way only, so there are no cycles.
- :User Management and :Payment Management depend on no other module.
- :Security & API calls all five modules (Figure 1).
- “read only”: Reporting only reads; it never changes other modules' data.

Legend

Component (one business module)

Dependency: the module at the tail calls the interface of the module it points to